

Lesson 10.2: Heuristics and biases

Psychology of decision-making

Victor Jauregui

Heuristics and bias

Definition (Correctness)

A judgement is *correct* if it affirms a true proposition or denies a false one. An *error in judgement* occurs when a judgement is incorrect.

Definition (Heuristic)

A *heuristic* is a (simplified) method of making judgements.

Definition (Bias)

A *bias* is a systematic error in judgement.

Some heuristics and their biases

Heuristics and probability judgements:

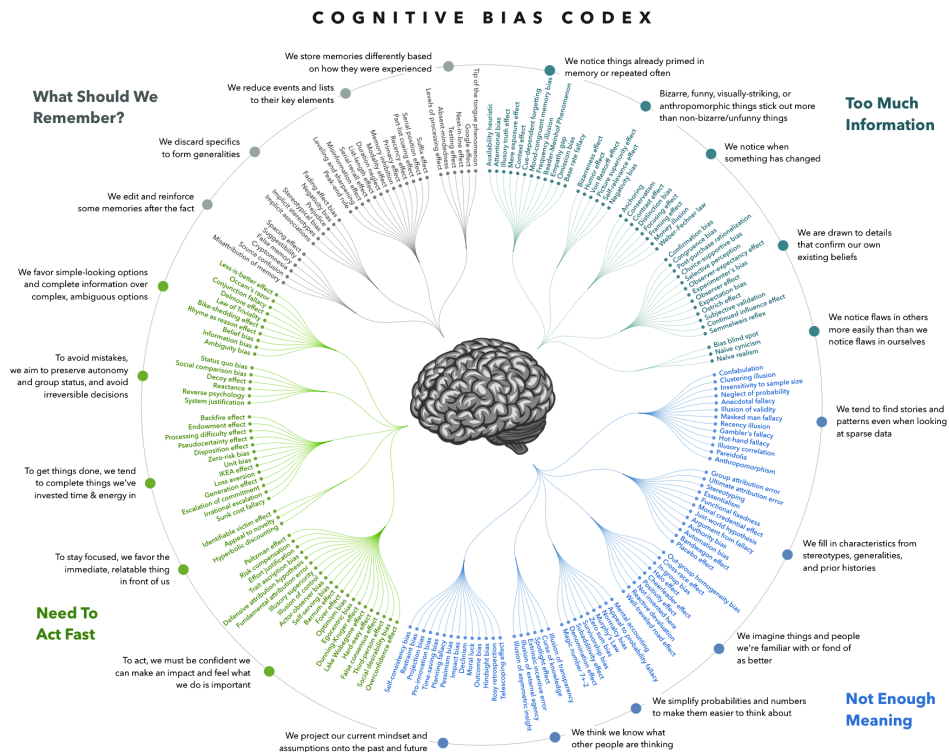
availability we judge events that come to mind more easily to be more likely; eg, it is easier to bring to mind memorable events (eg, terrorism), even if rare

representativeness we use similarity to archetypes to make classification judgements; eg, shy people resemble (stereo-)typical librarians more than farmers

Heuristics often substitute 'easier' propositions:

“shy X more likely to be a librarian than a farmer” by “shyness more typical of librarians than farmers”

Cognitive biases



Representativeness and probability judgements

From Kahneman, *Thinking, Fast and Slow*.

An individual has been described by a neighbor as follows: "Steve is very shy and withdrawn, invariably helpful but with little interest in people or in the world of reality. A meek and tidy soul, he has a need for order and structure, and a passion for detail."

Is Steve more likely to be a librarian or a farmer?

Kahneman attributes this to the *representativeness heuristic*

Decision-theoretic interpretation: Laplacian assumption

Laplace: ignorance = equal probability

